

Random agent :-

Batch 05

s_t = state at time t

A_t = Action selected at time t

R_{t+1} = Reward received

S_{t+1} = Next state

$$s_t \rightarrow A_t \rightarrow R_{t+1}, s_{t+1}$$

Random action selection

$$P(A_t = a) = 1/n$$

$\therefore$ Cast poker:

$$A = \{0, 1\}$$

$$P(0) = P(1) = 1/2$$

mountain cat:

$$A = \{0, 1, 2\}$$

$$P(0) = P(1) = P(2) = 1/3$$

Pao zen lake

$$A = \{0, 1, 2, 3\}$$

$$P(0) = P(1) = P(2) = P(3) = 1/4$$

Total reward

$$R_{total} = \sum_{t=0}^{T-1} R_{t+1}$$

$$\text{avg reward } R_{avg} = \frac{R_{total}}{T}$$

$$\text{Ex } R_1 = 1, R_2 = 1, R_3 = 1, R_4 = 1$$

Then

$$T = 4$$

$$R_{total} = 1 + 1 + 1 + 1 = 4$$

$$R_{avg} = 4/4 = 1$$

$$s_t \xrightarrow{A_t} s_{t+1}, R_{t+1}$$

Repeat until :-

Terminated = false.